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hetGPy: Heteroskedastic Gaussian Process Modeling in Python

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Summary

Computer experiments are ubiquitous in the physical and social sciences. When experiments are time-consuming to run, emulator (or surrogate) models are often used to map the input–output response surface of the simulator, treating it as a black box. The workhorse function of emulator models is Gaussian process regression (GPR). GPs provide flexible, non-linear regression targets with good interpolation properties and uncertainty quantification. However, it is well-known that naïve GPR scales cubically with input size, and specifically involves intensive computation of matrix determinants and solving linear systems when fitting hyperparameters (Garnett, 2023; Gramacy, 2020). Further, naïve GPR with noisy observations typically assumes an independent, identically-distributed noise process, but many data-generating mechanisms, especially those found in stochastic computer simulation, exhibit non-equal variance noise (also known as heteroskedasticity) (Baker et al., 2020). The software package `hetGP` (Binois & Gramacy, 2021) alleviates both of these concerns when the dataset of interest contains multiple observations per location (i.e. a dataset of size N has $n << N$ unique input locations). The `hetGP` framework can calculate matrix inverses and determinants (and therefore the log likelihood and its gradients) on the order $O(n^3)$ rather than $O(N^3)$ time in the naïve case (Binois et al., 2018). This is accomplished via multiple applications of the Woodbury matrix identity (Binois et al., 2018; Harville, 1997) and leveraging the replication structure. Further, `hetGP` also models the mean and noise processes as two coupled GPs, allowing smooth noise dynamics over parameter space (Binois et al., 2018; Gramacy, 2020, 2020). The package has been used in a variety of contexts, such as statistics (Binois et al., 2018, 2019), biology (Lazaridis et al., 2022) and computational epidemiology (Shattock et al., 2022). We present a Python reimplementation `hetGPy`, developed in part due to Python's widespread use in computer simulation (Downey, 2023; Kinser, 2022).

Statement of Need

Python is a popular language for software development, data science, and computer experimentation. Its object orientated framework, high-level functionality, and third-party libraries such as `numpy` (Harris et al., 2020), `scipy` (Virtanen et al., 2020), `PyTorch` (Paszke et al., 2019) and `scikit-learn` (Pedregosa et al., 2011) make it a powerful tool for academic and industry professionals alike. Python is especially popular for computer simulation, with one particular example being the widespread use of Python models of COVID-19 spread (Aylett-Bullock et al., 2021; Kerr et al., 2021; O'Gara et al., 2023). `hetGPy` is well-posed for sequential design of Python models, mirroring the functionality of hetGP without having to rely on intermediate

libraries such as `reticulate` (Ushey et al., 2023) or `rpy2`(Gautier, 2008), or in a more laborious case, converting a Python simulation to R.

The state of the art for GPR in Python is `GPyTorch` (Gardner et al., 2018), facilitated by black-box matrix–matrix multiplication (BBMM) which is extremely computationally efficient on GPUs. Other GPR routines for Python exist as well and can be found in libraries such as `PyMC` (Abril-Pla et al., 2023), `GPflow` (Matthews et al., 2017), and `GPJax` (Pinder & Dodd, 2022). However, while it is possible to jointly model the mean and variance as coupled GPs with these libraries, they do not take advantage of replication in datasets, meaning that under large degrees of replication and input-dependent noise, as is common in stochastic computer experiments (Baker et al., 2020), `hetGPy` will be more computationally efficient. Figure 1 (a) shows a heteroskedastic GP fit to a simulated motorcycle accident dataset (Silverman, 1985). While it is possible to model input-dependent noise in `GPyTorch` or `BoTorch` (Balandat et al., 2020), specifying a smooth noise process in the method of (Binois et al., 2018) would require a custom implementation. Figure 1 (b) illustrates the results of a simple one-dimensional example where we compare the model fits using both `hetGPy` and `GPyTorch`. While both models result in similar predictions, `hetGPy` is far faster, performing exact inference in less than 1 second, while GPyTorch takes over 10 seconds.

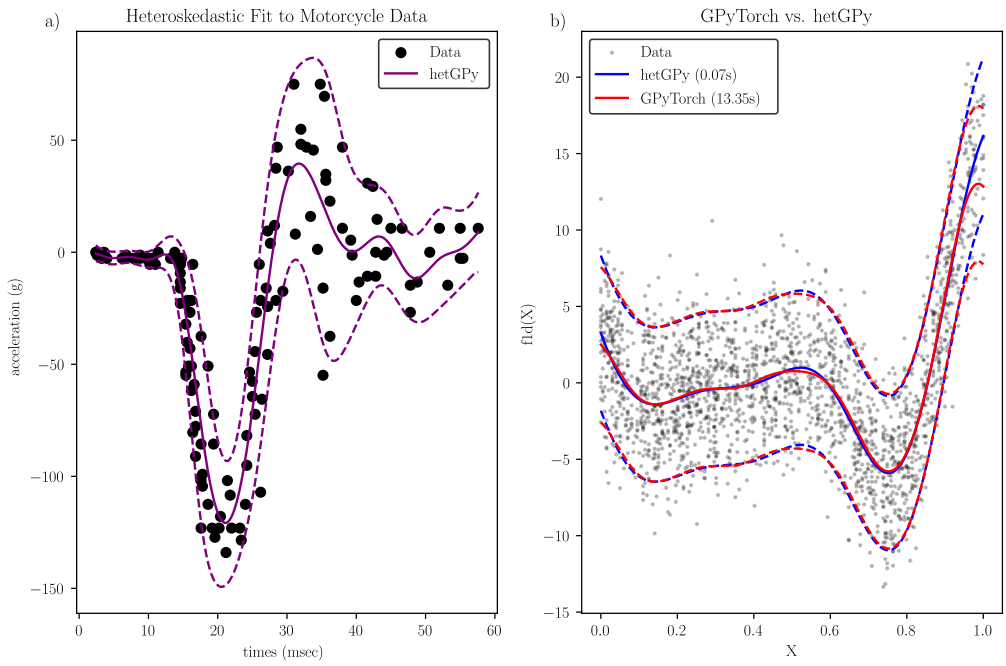

Figure 1: The main features of `hetGPy`. Panel (a) shows a heteroskedastic fit to the motorcycle data (Silverman, 1985). Panel (b) shows that `hetGPy` yields faster training than (naïve) `GPyTorch` with similar performance under high replication. Bolded and dashed lines indicate the predictive mean and 90% predictive intervals for homoskedastic GPR, respectively. Data were sampled from the f1d function $f(x) = (6x-2)^2 \sin(12x-4)$ (Forrester et al., 2008) in `hetGP` with between 1 and 20 replicates at each design location. Model training times in seconds are next to legend labels.

hetGPy also has two intermediate goals: (1) efficient computations, accomplished via implementation on numpy arrays and (2) minimal dependencies, the core of which are `numpy` for efficient array-based computation and `scipy` which contains the definitive implementation of the L-BFGS-B algorithm in Fortran (Byrd et al., 1995; Morales & Nocedal, 2011) used for maximum likelihood estimation of hyperparameters. Our experiments indicate `hetGPy` is able to learn response surfaces efficiently, and in the case of high replication, do so on CPUs more efficiently than the default implementation in `GPyTorch`, as shown for a suite of test problems

in Table 1. The test problems contain 1,000 unique design locations and an average of 5 replications per location, meaning $n = 1,000$ and $N \approx 5,000$ which shows why `hetGPy` is far more efficient in this setting. As a comparator, we also conduct a set of experiments using stochastic kriging (SK) (Ankenman et al., 2010), a precursor method to Binois et al. (2018) that also allows for maximum likelihood estimation with replication, and under the case of homoskedasticity, is nearly equivalent to homoscedastic GPR in `hetGP` and `hetGPy` (Gramacy, 2020). We implement SK with a custom `GPyTorch` likelihood that accounts for replication under homoskedasticity. Specifically, given a dataset X with unique designs $(X_1, ..., X_k)$ each replicated $(n_1, ..., n_k)$ times, we pre-average the outputs Y_i at each unique input location, and then estimate the diagonal of the noise matrix as $(\sigma^2 \,/\, n_i)$. We see that for the SK case, `hetGPy` and `GPyTorch` have training times on a similar order of magnitude. The package `hetGPy` is under active development and is well-posed to engage with the wider Python community for future extension such as arbitrary kernel functions with auto-differentiation methods facilitated by `PyTorch` or `jax` (Bradbury et al., 2018).

		Time (Seconds)			Number of Function Evaluations		
Covariance	Experiment	GPyTorch (naïve)	GPyTorch (SK)	hetGPy	GPyTorch (naïve)	GPyTorch (SK)	hetGPy
Gaussian	Branin	98.45	5.61	2.96	13	18	13
	Goldstein-Price	119.02	3.7	2.04	21	12	10
	Hartmann-4D	98.57	4.67	5.15	18	15	16
	Hartmann-6D	466.38	21.05	21.49	95	69	40
	Sphere-6D	206.05	14.54	7.83	42	51	20
Matern $\nu = 5/2$	Branin	90.5	4.88	6.62	13	12	20
	Goldstein-Price	153.03	4.55	4.0	20	11	15
	Hartmann-4D	201.21	7.24	6.73	21	18	16
	Hartmann-6D	339.39	29.45	45.78	53	67	57
	Sphere-6D	147.12	16.89	16.46	24	38	14

Table 1: Comparing training times across a suite of test problems and libraries. All experiments reflect exact GPR with homoscedastic noise. Optimization problems were selected from (Picheny et al., 2013) with implementations from (Surjanovic & Bingham, 2013). Experiments consisted of a Latin Hypercube design of 1,000 unique locations, with between 1 and 10 replicates. iid Gaussian noise was added to each resulting dataset. Experiments were conducted on a 2019 MacBook Pro with a 2.6 GHz 6-Core Intel Core i7 processor, 16GB RAM on macOS Sonoma 14.7.1.

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